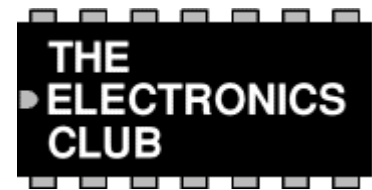


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'Random' Flasher for 8 LEDs Project



A kit for this project is available from [RSH Electronics](#).

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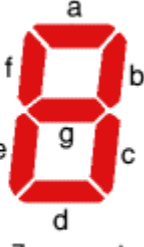
This project flashes eight LEDs in an apparently random manner. It uses a 4026 combined counter and display driver IC which is designed for driving 7-segment LED displays. The sequence is not really random because seven of the LEDs would normally be the display segments, the eighth LED is driven by an output that is normally used for driving further counters. The table below shows the sequence for the LEDs. You can use less than eight LEDs if you wish and the table may help you decide which ones to use for your purpose.

This project uses a [555 astable](#) circuit to provide the clock pulses for the [4026 counter](#). Resistors are not required for the LEDs because the 4026 IC limits the current to about 5mA for each LED.

Parts Required

- resistors: 10k, 47k
- capacitor: 1µF 16V radial
- 3mm or 5mm LEDs ×8, they can be various colours
- 555 timer IC
- 4026 counter and display driver IC
- DIL sockets for ICs: 8-pin, 16-pin
- battery clip for 9V PP3
- stripboard: 16 rows × 14 holes

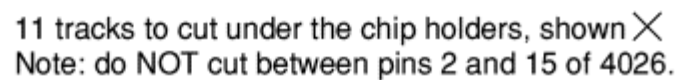
Outputs from the 4026 counter and display driver IC								
Count	a	b	c	d	e	f	g	h
0	•	•	•	•	•	•		•
1		•	•					•
2	•	•		•	•		•	•
3	•	•	•	•			•	•
4		•	•			•	•	•
5	•		•	•		•	•	
6	•		•	•	•	•		
7	•	•	•					
8	•	•	•	•	•	•	•	
9	•	•	•	•		•	•	



7-segment display

• = segment on. h is used to drive other counters.

Stripboard Layout



The diagram shows a 555 timer (IC 1) and a 4026 counter and display driver (IC 2). The 555 timer is configured with a 10k resistor and a 47k resistor connected to its pin 7, and a 1μF capacitor connected to its pin 1. The 4026 counter and display driver is connected to the 555 timer's output (pin 3) and its pin 1. The 4026 driver has seven outputs (pins 5, 6, 7, 9, 10, 11, 12) connected to seven LEDs labeled h, f, g, d, a, e, and b. The LEDs are connected to a common ground.



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